



Scientific Working Group on Digital Evidence

Core Competencies for Mobile Device Forensics

SWGDE 12-F-003-2.0

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1. Purpose

This document provides an outline of the knowledge and abilities all practitioners of mobile device forensics should possess. The following elements provide a basis for training and testing programs. This basis is suitable for authorization, competency assessment, and proficiency testing.

2. Scope

This document identifies the core competencies necessary for the handling and forensic processing of mobile devices through the life cycle of an investigation. This document applies to anyone involved in any of these tasks. For the purposes of this document, the term “examiner” refers to individuals who have specialized training, knowledge, skills, and abilities that allow them to handle a wide range of technical issues related to mobile device forensics, and who may be performing technical tasks to include collection, acquisition, analysis, and reporting.

Not all core competencies will be relevant to every practitioner’s role in a forensic services organization. These organizations must determine which core competencies are within the scope of their organization and examiners. Lack of competence in one component may not invalidate overall competency.

There is a spectrum of capabilities within core competencies. It is not expected that an examiner has to be proficient in every capability to be considered competent. Examiners should exhibit competence pertinent to the examination being undertaken.

This document does not address core competencies for chip-off or micro-read analysis.

Refer to *SWGDE 10-Q-002-3.0 Guidelines & Recommendations for Training in Digital & Multimedia Evidence* for general training requirements of forensic practitioners.

An examiner should apply all principles as defined in *SWGDE 10-Q-001-1.0 Minimum Requirements for Quality Assurance in the Processing of Digital and Multimedia Evidence*.

3. Limitations

This document is not all-inclusive, does not contain information relative to or in support of specific commercial products, and is not intended to be a training manual or to specify operating procedures.

4. General Considerations

Examiners engaging in mobile device forensics activities should be confirmed by their organization to meet criteria such as capabilities, education, training history, certification, competency assessment(s), and final authorization determined by the organization to carry out examinations.

Mobile device forensics continues to be a dynamic and specialized subdiscipline of digital



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forensics. As noted above, within an organization specialized roles may perform one or more tasks associated with the collection, acquisition, analysis, and reporting of mobile devices.

Personnel conducting these tasks should be trained to the competencies relevant to their tasks and should remain current in developments in mobile device technologies. This may be accomplished by reading trade journals, taking classes, participating in professional organizations, taking continuing education, on the job training, and hands-on experience.

5. Mobile Device Forensics Core Competencies

An examiner must be able to recognize circumstances beyond their expertise and seek appropriate guidance, consulting with specialists as needed. The categories of core competencies are as follows:

- Legal Considerations and Ethical Standards
- Foundational Skills
- Evidence Identification
- Collection, Seizure, and Preservation
- Data Acquisition
- Examination and Analysis
- Documentation
- Presentation and Testimony

5.1 Legal Considerations and Ethical Standards

- Sufficient training to understand and apply authorization to conduct a search and seizure of digital devices (e.g., the ability to read a search warrant and determine scope including what places may be searched and what data may be seized)
- Awareness and understanding of applicable laws and organizational policies relevant to handling digital evidence or computer-related crimes
- Understanding jurisdictional differences, informed by local, state, and federal guidelines, related to digital evidence and privacy regulations
- Understand the Stored Communications Act and its implication to mobile device forensics.
- Adherence to ethical guidelines and professional standards established by professional organizations within the digital forensics community to ensure impartiality, proportionality, confidentiality, and legal compliance in the collection, analysis, and reporting of digital evidence
- Understanding of how cognitive bias may influence the collection, perception, and interpretation of data so as to minimize its impact to the digital forensic process
- Balancing the need to review relevant evidence while minimizing intrusion into privacy

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5.2 Foundational Skills

- Ability to describe how mobile devices operate and communicate
- Knowledge of how and when to use Personal Protective Equipment (PPE)
- Knowledge of what equipment may be needed for both on-scene or lab examinations (e.g., cables, drives, camera, software)
- Knowledge of organizational policies, procedures, and best practices regarding the collection and analysis of digital evidence
- Understanding the importance of forensic tool testing and validation to establish confidence in forensic results
- Ability to prepare a forensic workstation for use during forensic examinations
- Understanding of foundational mobile device forensic concepts including:
 - Reading and converting numbering systems relevant to computing (e.g., hexadecimal, binary)
 - Wiping/sterilization of media
 - Mobile device file systems and logical structure
 - Compound files
 - Encoding/Decoding
 - File signatures
 - Parsing
 - Metadata
 - Keyword searching and search expressions
 - Encryption/Decryption
 - Common database file formats on mobile devices
 - Cloud storage and backups
 - Use of hash algorithms
 - Mobile device components and their functions
 - Common application data storage techniques
 - Device recovery states
 - Mobile device security technologies (e.g., secure enclave, secure startup)
 - Device states and modes of operation

5.3 Evidence Identification

- Ability to identify mobile devices
- Understand the differences between physical SIM cards and eSims and identify devices that potentially utilize one or multiple SIMs
- Ability to differentiate between handset lock, PIN lock, and PUK

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- Ability to locate/identify any potentially applicable removable storage media and/or peripheral devices (e.g., watches, Bluetooth devices)
- Ability to identify the type of information that may be stored in the device and each storage media available to the device
- Understanding differences that may be encountered when processing removable media while in the device versus processing it externally
- Understanding that data from devices or removable media may not be acquired by tools in all instances
- Understanding the differences between static and volatile data sources
- Ability to recognize volatile data and the access-state of various devices (on/off/locked/unlocked) and respond according to best practices for data access and integrity
- Ability to locate device identifiers (e.g., ESN, MEID, IMEI, FCC ID, MIN, SID)
- Knowledge of the various types of identity cards (e.g., SIM, USIM, CSIM, RUIM)
- Ability to locate SIM card identifiers when present (e.g., IMSI, ICCID, MSIN, MSISDN).
- Knowledge of physical characteristics of various SIM card sizes (e.g., standard, mini, micro, and nano)

5.4 Collection, Seizure, and Preservation

- Understand the possible need to process the device for other forensic evidence prior to extracting its data (e.g., fingerprints/DNA/blood/trace evidence issues)
- Ability to execute a planned collection process to establish chain of custody for seized items and maintain quality control according to organizational guidelines
- Ability to practice general collection safety and determine the best method of collection to preserve maximum information relevant to the incident or case
- Understand the consequences and risks associated with previewing a mobile device prior to data acquisition
- Understand the impact of battery depletion, removal, or replacement
- Ability to effectively communicate with investigative team to obtain pertinent information to possibly access a device
- Understanding of digital evidence packaging during collection and storage to protect the physical device from environmental damage
- Identify the steps necessary to maintain the integrity of removable media (e.g., SIM cards or memory cards)
- Understand the proper way to decontaminate a mobile device damaged by fluids (e.g., water or bodily fluids)
- Ability to apply appropriate radio and network isolation techniques for a given mobile device (e.g., Airplane Mode, Faraday bag)

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- Ability to explain the potential outcome of powering off or restarting a mobile device, taking device specific considerations into account
- Understand how to recognize and protect devices that are physically damaged
- Understand how to repair minor damage to mobile devices (e.g., damaged screens, part replacement)
- Ability to ensure device has sufficient battery charge or externally supplied power to complete the data extraction

For additional information, see *SWGDE 18-F-003-1.2 Best Practices for Mobile Device Evidence Collection & Preservation Handling and Acquisition*.

5.5 Acquisition

- Understand the different types of acquisitions, tool limitations and the possible need for additional acquisitions (e.g., logical extraction of data may not retrieve deleted data from the handset, SIM card or memory cards)
- Ability to demonstrate the use of forensic tools to acquire a device image and validate the integrity of the resulting image through hash verification
- Ability to troubleshoot software as well as physical hardware to the extent required for acquisition and processing
- Ability to identify appropriate forensic tool for acquisition of data from devices
- Ability to conduct risk assessments for issues that may arise when using these tools (e.g., jailbreak, APK downgrade)
- Knowledge of tool functionality, the limitations and the possible need for additional acquisitions (e.g., logical extraction of data may not retrieve deleted data from the handset, SIM card or memory cards)
- Understand when it may be necessary to conduct a manual analysis of a mobile device
- Understand how processing a mobile device can possibly alter the data contained within the device
- Understand the different connectivity options utilized during data acquisition (Cable/Bluetooth)

For additional information, see *SWGDE 18-F-003-1.2 Best Practices for Mobile Device Evidence Collection & Preservation Handling and Acquisition*.

5.6 Examination and Analysis

- Knowledge of how to verify data integrity through the use of hash validation



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- Basic recognition and understanding of common mobile device operating systems (e.g., Android, iOS)
- Ability to recognize and acquire data from various commonly utilized file system formats
- Ability to determine the appropriate tool(s) for the forensic task being performed
- Ability to recognize commonly installed applications with encrypted data
- Knowledge of the type of backup files used by mobile devices and where to locate those files on the computers synced to the mobile device
- Understanding of the types and locations of data stored on SIM cards (e.g., Service-related information—ICCID, IMSI, MSISDN, SPN; Phonebook and call information—abbreviated and last dialed numbers; Messaging information—SMS, EMS; Location information—LOCI, GPRSLOCI)
- Ability to understand various methods of record deletion and their impact to recoverability

For additional information, see *SWGDE 18-F-003-1.2 Best Practices for Mobile Device Evidence Collection & Preservation Handling and Acquisition*.

5.7 Documentation

- Ability to record contemporaneous notes while conducting the acquisition and examination
- Ability to locate software audit logs to ensure repeatability and reproducibility
- Ability to write reports per your organizational guidelines containing all relevant information in a clear and concise manner to include unique identifiers of digital device(s) and software versions of forensic tools used
- General photography skills to document physical condition, manual analysis, and evidence on site or on the target media

For additional information, see *SWGDE 18-Q-002-1.0 Requirements for Report Writing in Digital and Multimedia Forensics*.

5.8 Presentation and Testimony

- Ability to develop demonstrative exhibits for legal proceedings
- Ability to articulate the types and use of forensic tools during the acquisition process (e.g., how the tools operate, any issues during the acquisition)
- Ability to present technical findings clearly and concisely to a non-technical audience

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For additional information, see *SWGDE 23-Q-001-1.0 Best Practices for Personnel Presenting Digital Evidence in Legal Proceedings* and *SWGDE 22-Q-001-1.1 Introduction to Testimony in Digital and Multimedia Forensics*.

6. Additional Resources

- Scientific Working Group on Digital Evidence. *Best Practices for Mobile Device Evidence Collection & Preservation Handling and Acquisition*. SWGDE 18-F-003-1.2. SWGDE, 2018, <https://www.swgde.org/18-f-003/>.
- Scientific Working Group on Digital Evidence. *Best Practices for Personnel Presenting Digital Evidence in Legal Proceedings*. SWGDE 23-Q-001-1.1. SWGDE, 2023, <https://www.swgde.org/23-q-001/>.
- Scientific Working Group on Digital Evidence. *Guidelines & Recommendations for Training in Digital & Multimedia Evidence*. SWGDE 10-Q-002-3.0. SWGDE, 2010, <https://www.swgde.org/10-q-002/>.
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7. History

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1.0 DRAFT	6/5/2012	Initial draft created. SWGDE voted to release as a Draft for Public Comment. Formatted for release as a draft for public comment.
1.0 DRAFT	1/17/2013	Addressed public comments received.
1.0	2/11/2013	SWGDE voted to publish as Final Approved document. Formatted for Final Approved document.
1.0	9/27/2014	Document updated per current SWGDE policy with new disclaimer. Removed the Definitions section, and corrected SWGDE hyperlinks. No changes to content and no version/publication date change.
2.0 DRAFT	1/16/2025	Significant reformatting and information update. SWGDE voted to release as a Draft for Public Comment.
2.0 DRAFT	2/10/2025	Formatted for release as a draft for public comment.
2.0 DRAFT	5/22/2025	No public comments received. Moved forward for SWGDE vote to publish as a Final Approved Document.
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